# How AI Reshapes Human Content Creation: The Case of Wikipedia

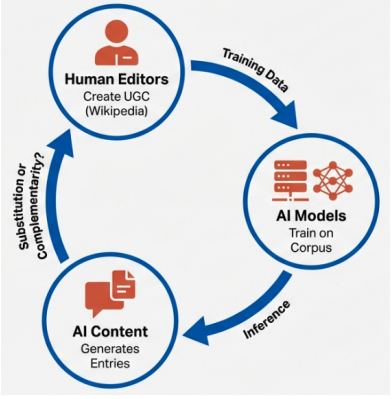
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CREED

# Can AI Compete with Human Knowledge Production?

- Wikipedia is produced by **volunteer editors**.
- Human-created knowledge supplies **training data for AI**.
- AI-generated content can compete with these same sources.

Does AI crowd out human contribution, or stimulate it?



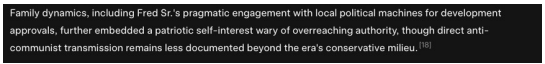
# Donald Trump: An Illustration of the Inspiration Channel

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## Grokipedia: October 27


Family dynamics, including Fred Sr.'s pragmatic engagement with local political machines for development approvals, further embedded a patriotic self-interest wary of overreaching authority, though direct anti-communist transmission remains less documented beyond the era's conservative milieu.[18]

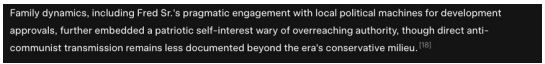
Launch-day discussion of Trump's background:

"...direct **anti-communist transmission** remains less documented..."

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## Wikipedia revision: October 30

**Revision as of 00:51, 30 October 2025 (view source)**
NesserWiki (talk | contribs)
*(→External links: Added to Category:American anti-communists)*
*(Tags: Reverted. Mobile edit. Mobile web edit. Advanced mobile edit)*

An editor adds the category **"American Anti-Communists"**.

External AI content can draw attention to omissions or alternative framing.

# Two Channels Affect Human Editing

## Views / attention

$$\text{Views}_i = \bar{V}_i + G_i(\theta - \delta s_i)$$

- AI can attract attention to topics or substitute for Wikipedia readership.
- Larger Wikipedia audiences encourage editing.

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## Inspiration / information differences

$$b_i = \kappa\,\text{Views}_i + G_i\rho(1 - s_i)$$

- Different information or framing reveals possible omissions.
- Editors investigate and revise Wikipedia.

**Prediction:** lower similarity predicts more editing among covered pages.
Both channels can increase editing and operate together.

# Wikipedia and Grokipedia Data

## Pages, activity, and text

- **1.4 million** English Wikipedia biographies in five fields.
- About **170,000** Grokipedia entries identified during collection.
- Page views and edits; editor identities and article text.

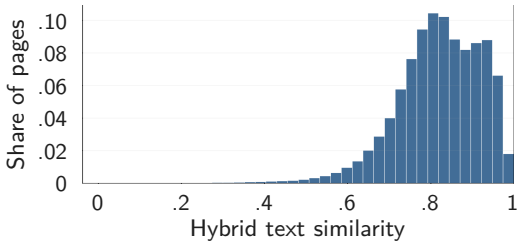
Academia, Culture, Leaders, Politics, and Sports.

**Launch: October 27, 2025**

8 pre-launch weeks: Sep. 1–Oct. 26

8 post-launch weeks: Oct. 27–Dec. 21

## Initial textual similarity

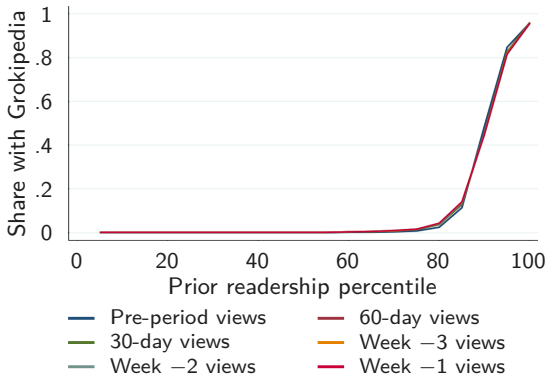



**80,687** matched treated pages; **7.9%** above 0.95.

0.6 transformer semantic + 0.4 TF–IDF lexical.

Initial differences may predict editing responses.

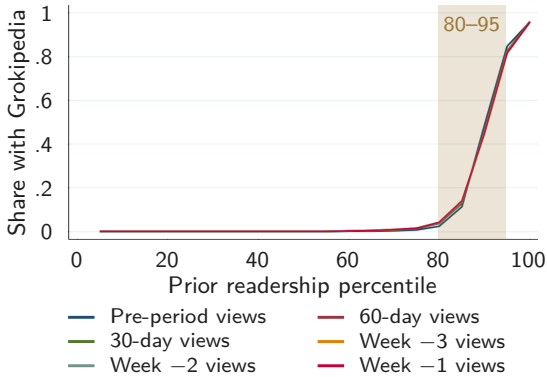
# Matching Creates Comparable Pre-Launch Histories




## Selection into Grokipedia

Coverage rises with prior readership.

# Matching Creates Comparable Pre-Launch Histories

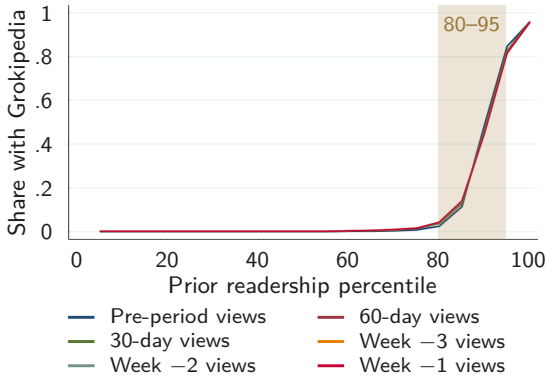



## Selection into Grokipedia

Coverage rises with prior readership.

**80–95:** covered and uncovered pages can have similar popularity.

# Matching Creates Comparable Pre-Launch Histories




**Assumption:** the matched gap would stay stable without Grokipedia.

## Selection into Grokipedia

Coverage rises with prior readership.

**80–95:** covered and uncovered pages can have similar popularity.

## Match, then compare changes

- Within fields: eight-week views and **exact weekly total-edit profiles**.
- Historical readership, length and edits; a mean-view caliper.

**80,909** treated–control pairs.

DiD compares pre-to-post changes.

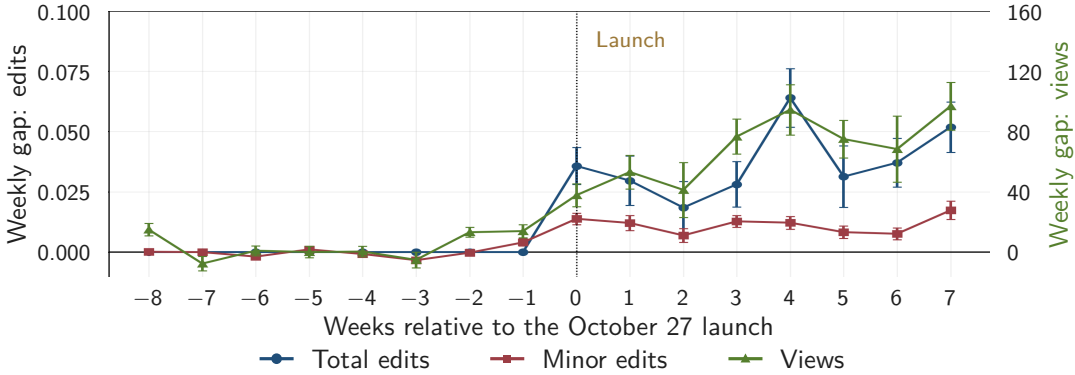
# Wikipedia Views and Edits Increase

**Eight-week total-count DiD per page**, relative to matched controls

| | DiD estimate | Pre-launch baseline | % of baseline |
|---|---|---|---|
| Page views | +512.94 | 3,172.38 | **16.2%** |
| Total edits | +0.296 | 0.637 | **46.5%** |
| Minor edits | +0.092 | 0.187 | **49.3%** |

The editing response extends beyond edits marked as minor.

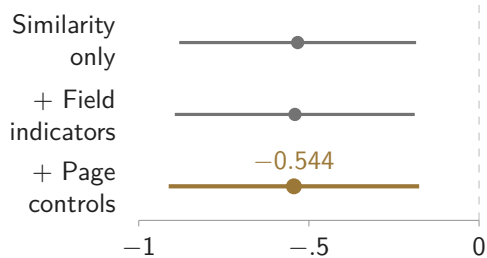
# Positive Weekly Gaps Persist After Launch




Pre-launch total-edit gaps are zero **by matching**, not an independent parallel-trends test.

# Lower Initial Similarity Predicts More Editing

**Similarity coefficient on total-edit DiD**



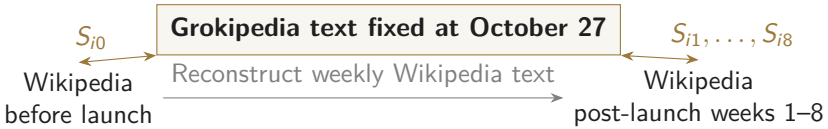

95% confidence intervals

A **0.10 reduction** in similarity is associated with **+0.054** additional total edits over eight weeks.

Different information or framing offers more prompts for revision.

Consistent with **inspiration**. Similarity is not randomly assigned.

# Initially Distant Pages Show Greater Relative Convergence




## Larger initial gaps

More positive eight-week changes in similarity to the fixed AI text.

**+0.00636** initial-gap coefficient

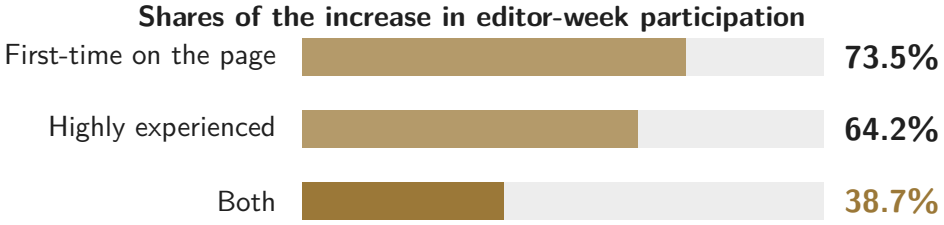
## More post-launch editing

The positive convergence gradient becomes stronger where editing is greater.

**+0.02021** gap × editing

Relative convergence is consistent with inspiration rather than pure differentiation.

# Experienced Editors Contribute to Additional Pages




**First-time on the page:** first observed contribution to that page, not a new Wikipedian.

**Highly experienced:** at least 1,000 English-Wikipedia revisions before September 1, 2025.

# AI Can Complement Human Knowledge Production

1. Grokipedia's introduction is associated with **more Wikipedia readership and editing** relative to matched controls.
2. Initial content differences predict more editing and relatively greater **movement toward the AI text**.
3. Much of the participation response comes from **experienced editors entering additional pages**.

AI on a **competing platform** can redirect human attention and expertise.

Questions and discussion